

Scatter Plots and Informal Lines of Fit

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

1. 78	4. 55	7. 5.1	10. 76.5
2. 93	5. 97	8. -0.5	11. 37, 12
3. 0.7	6. 6.6	9. 5.5, 5.7	12. 195

Curriculum

Level: Grade 8 / Pre-Algebra

8.SP.A.1 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–5 of 5 across 20 tagged checks.

1. Revision data: what score did the student who revised for 30 minutes earn?

Solution. Find 30 on the horizontal axis, go straight up to the point, then straight across to the vertical axis. The point is (30, 78), so the score is 78 points. Reading across first and reporting 30 is the standard slip — that is the input, not the score.

78

2. Describe the association in the revision data, and give the highest quiz score.

Solution. As revision time increases the scores increase, so the association is POSITIVE. The points lie close to a single straight path with no large gaps, so it is FAIRLY STRONG and roughly LINEAR. There is no clustering into separate groups and no outlier. Scanning the vertical coordinates, the largest is at (55, 93), so the highest score shown is 93 points.

93

3. Find the slope of the line of fit through (10, 62) and (50, 90).

Solution. A line of fit is drawn by eye through the middle of the cloud, with roughly as many points above as below — it does not have to touch any point. Using the two named points,

$$m = \frac{90 - 62}{50 - 10} = \frac{28}{40}.$$

In context, the slope means about 0.7 of a point of score for each extra minute of revision.

 $m = 0.7$

4. The line passes through (10, 62) with slope 0.7. Find b and write the equation.

Solution. Substitute the point into $y = mx + b$: $62 = 0.7(10) + b$, so $62 = 7 + b$ and $b = 62 - 7$. The equation of the line of fit is $y = 0.7x + 55$. The intercept 55 is the score the line predicts for zero minutes of revision — plausible here, but it is a prediction outside the data and should not be leaned on.

 $b = 55$

5. Predict the score for 60 minutes using $y = 0.7x + 55$.

Solution. Substitute $x = 60$: $y = 0.7(60) + 55 = 42 + 55$. The predicted score is 97 points. The data runs from 10 to 55 minutes, so 60 sits just past the top of the range: this is a short EXTRAPOLATION, close enough to the data to be reasonable. A student who stops at 42 has used the slope term and forgotten to add the intercept.

97

6. Screen-time data: how much sleep did the student with 5 hours of screen time get?

Solution. The point above 5 on the horizontal axis sits at height 6.6, so that student slept 6.6 hours.

6.6

7. Describe the association in the screen-time data, and give the least amount of sleep shown.

Solution. As screen time increases, sleep decreases, so the association is NEGATIVE, and the points fall in a fairly tight straight band, so it is reasonably strong and linear. The lowest point on the plot is (8, 5.1), so the least sleep shown is 5.1 hours.

5.1

8. Line of fit through (0, 9) and (8, 5): slope, equation, and what the slope means.

Solution. The slope is

$$m = \frac{5 - 9}{8 - 0} = \frac{-4}{8},$$

so $m = -0.5$. Since the line crosses the vertical axis at (0, 9), the intercept is $b = 9$ and the equation is $y = -0.5x + 9$. In context: each extra hour of screen time is associated with about half an hour less sleep. Note the word ASSOCIATED — the line describes a pattern in this sample, it does not prove that screen time causes lost sleep.

 $m = -0.5$

9. Predict the sleep for 7 hours of screen time, and compare with the actual value.

Solution. Substituting into the line: $y = -0.5(7) + 9 = -3.5 + 9$, so the line predicts 5.5 hours. The plotted point for 7 hours of screen time is at 5.7 hours. The actual value is the larger one, by 0.2 of an hour (about 12 minutes). That gap is the residual — the vertical distance from the point to the line — and small residuals in both directions are exactly what a good line of fit produces.

5.5 predicted, 5.7 actual

10. Shoe size against quiz score: is a line of fit useful? Compute the mean quiz score.

Solution. The points scatter with no upward or downward drift: knowing a student's shoe size tells you nothing about the score. There is NO association, so no line of fit summarises this data — any line drawn here would have large residuals in both directions and would invite predictions that are not supported. What does describe the scores is their centre:

$$\frac{74 + 88 + 65 + 91 + 70 + 84 + 62 + 78}{8} = \frac{612}{8}.$$

The mean quiz score is 76.5 points.

76.5

11. Plant heights: name the outlier, give both ranges, and say how the outlier affects a line of fit.

Solution. The first seven points climb steadily by 2 cm a day, from 3 up to 15 cm. The eighth point, $(8, 40)$, sits far above that path — it is the outlier. Range of all eight heights: $40 - 3 = 37$ cm. Range of the seven without it: $15 - 3 = 12$ cm, a third as large. A line of fit drawn through all eight points is dragged upward at the right-hand end by that single point, so it comes out steeper than the other seven points support and it sits above most of them. Either exclude the outlier and say so, or check whether it is a measurement error.

37, 12

12. Predict the score for 200 minutes of revision from $y = 0.7x + 55$, and say why not to trust it.

Solution. Substituting: $y = 0.7(200) + 55 = 140 + 55$, so the line predicts 195 points. Two things are wrong with the prediction. First, 200 minutes is far outside the range of the data, which stops at 55 minutes — the line was fitted to that range and there is no evidence it continues. Second, the answer is impossible: a quiz is scored out of 100, so the real relationship must flatten out long before this. A line of fit is a summary of the data it was drawn through, not a law.

195
